

Neural Network Fitting Results

Inference evaluation of the MLP penetration surrogate on the clean split

Linan Jiang

Aalto University

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Evaluation Target

Source of this result set

The inference script loads the weights from the refined student model and predicts the penetration mean and predictive standard deviation point by point on the `series_wide_clean` trajectories.

Item	Setting
Refinement run	<code>distill_cdf_onset_v2_20260410_103413</code>
Teacher run	<code>stage2_engineered_nll_a_only_20260410_020607</code>
Evaluation split	<code>clean</code> , time window 0 ms to 5 ms
Input features	time, tilt angle, plume count, injection duration, control backpressure
Outputs	predicted penetration μ , predicted uncertainty σ , onset logit

Raw result folder: `MLP/inference_testing/rmse_eval_clean_20260415_092112`

Data Scale and Overall Error

Metric	Value
CSV files	266
Trajectories	14,972
Prediction points	399,738
Truth range	0.00 mm to 95.88 mm
RMSE	6.90 mm
MAE	5.16 mm
Bias	-1.29 mm
Median absolute error	3.97 mm
P90 / P95 absolute error	11.20 mm / 14.17 mm
Range-normalized RMSE	7.19%

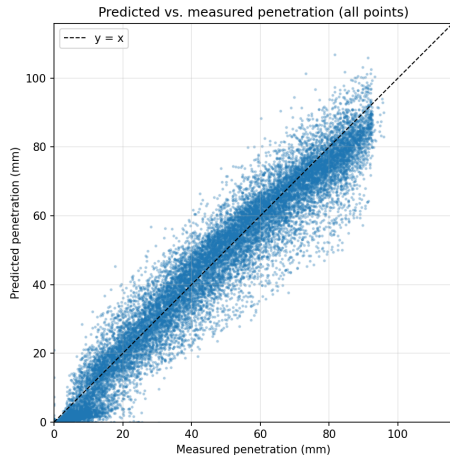
How to read the metrics

- **Typical error is modest:** the median absolute error is about 4 mm.
- **The global bias is negative:** on average, the model under-predicts penetration by about 1.3 mm.
- **Tail errors remain visible:** the P95 absolute error reaches 14.2 mm; a subset of conditions or trajectories still drives the main risk.

Uncertainty

- Mean predicted σ : 6.36 mm
- 1σ coverage: 68.47%
- 2σ coverage: 95.67%

Predicted vs. Measured Penetration

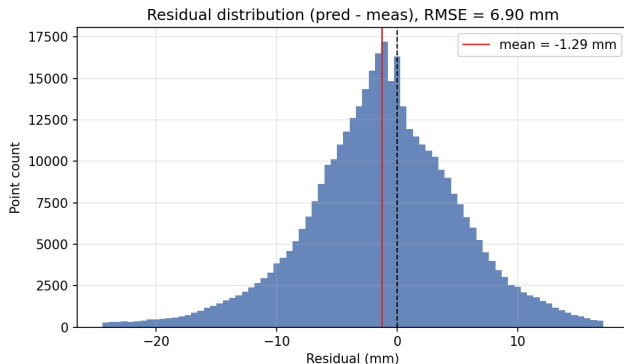


Main observations

- Most points follow the $y = x$ line, showing that the model captures the main penetration trend.
- Scatter increases in the high-penetration region, which is one of the main contributors to RMSE.
- A few nonzero predictions appear in the low-penetration region, reflecting timing and threshold uncertainty near onset.

Each point is one time sample. The axes are measured and predicted penetration.

Residual Distribution and Calibration

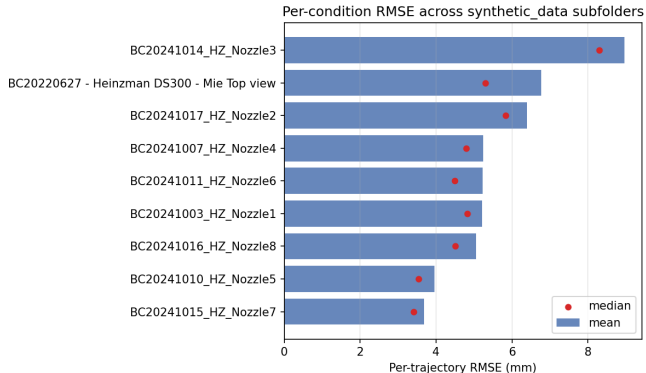


Residual definition

$$e = \hat{y}_{\text{pen}} - y_{\text{pen}}$$

- Bias = -1.29 mm, so the model is slightly biased toward under-prediction.
- 1σ coverage = 68.47%, close to the Gaussian one-sigma reference.
- 2σ coverage = 95.67%, close to the Gaussian two-sigma reference.
- Predicted σ is usable at the global scale, but it cannot replace condition-by-condition error analysis.

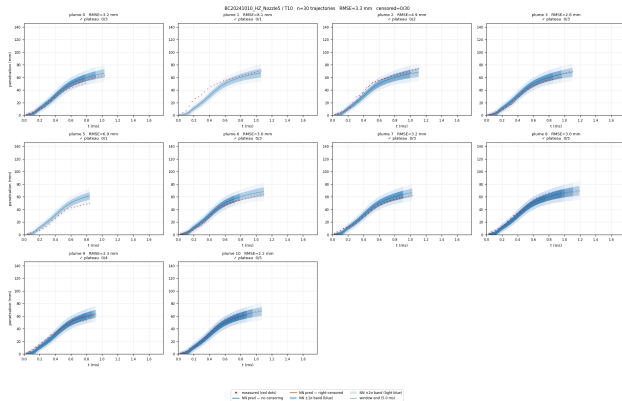
RMSE Grouped by Experimental Folder



Folder	Mean RMSE
Nozzle7	3.68 mm
Nozzle5	3.95 mm
Nozzle8	5.05 mm
Nozzle1	5.21 mm
Nozzle6	5.22 mm
Nozzle4	5.24 mm
Nozzle2	6.40 mm
DS300 top view	6.77 mm
Nozzle3	8.96 mm

- Best groups: Nozzle7 and Nozzle5, with RMSE below 4 mm.
- Hardest group: Nozzle3, with mean RMSE near 9 mm.

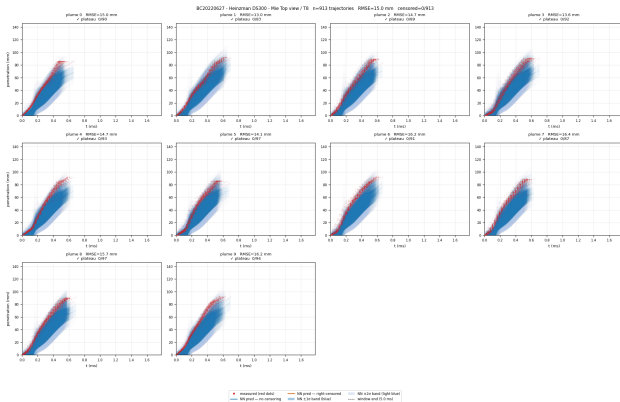
Well-Predicted Trajectory Group: Nozzle5 / T10



Representative characteristics

- Per-folder mean RMSE: 3.95 mm
- 1σ coverage: 73.65%
- 2σ coverage: 97.15%
- The growth trend is tracked consistently.
- The uncertainty band covers most plume/repeat trajectories.
- Engineered features and refinement capture the main condition dependence.

High-Error Trajectory Group: DS300 / T8



Risk signals

- DS300 folder mean RMSE: 6.77 mm
 - 1σ coverage: 60.00%
 - 2σ coverage: 92.98%
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- Worst cases concentrate in DS300/T8, with RMSE around 23.8 mm to 24.5 mm.
 - The low coverage indicates that uncertainty is still too narrow for outliers.
 - Raw segmentation, delay alignment, and condition distribution should be checked.

Conclusions

- ① **The overall fit is usable:** on the clean split, RMSE is 6.90 mm, NRMSE is 7.19%, and the median absolute error is 3.97 mm.
- ② **The uncertainty is reasonably calibrated overall:** $1\sigma/2\sigma$ coverage is 68.47%/95.67%, close to the Gaussian reference.
- ③ **The error is not uniformly distributed:** Nozzle5/7 perform well, while Nozzle3, DS300, and some high-penetration trajectories contribute most of the tail error.
- ④ **Next priorities:** perform segmentation QA, delay-alignment QA, and feature-coverage checks for high-error folders/tests, and evaluate whether folder-aware calibration is needed.

Short thesis-ready conclusion

The neural surrogate reproduces most penetration trajectories within the 0 ms to 5 ms window and provides predictive uncertainty that is reasonable at the global scale. Further improvement should focus on cross-dataset domain shift and a small number of high-error trajectories.

Content	File
Slides source	Thesis/neural_network_fit_results_slides_en.tex
Compiled PDF	Thesis/neural_network_fit_results_slides_en.pdf
Copied figures	Thesis/images/neural_network_fit_results/
Result folder	rmse_eval_clean_20260415_092112
Overall metrics	metrics_summary.json
Per-folder table	per_folder.csv
Per-trajectory table	per_trajectory.csv